

Research Notes

1 Introduction

Document will outline my approach to learning hydraulics and how to create a hydraulic launcher. Hydraulic launchers are interesting because they are used often in rollercoaster launch systems, see figure 1, military drones, and aircrafts. Often, large drones are catapulted with hydraulic launchers because of their ability to deliver high force, and precise speed in a compact area.



Figure 1: *Wicked* rollercoaster from theme park in Utah [1].

I'll detail in this report the exact mechanics of hydraulics and how one can fabricate one from scratch.

2 Hydraulics

Hydraulics move heavy machinery, like construction equipment, around using trapped liquid. That might be ambiguous, but that trapped liquid compresses and exerts massive forces due to pressure. A good example is taking an empty syringe, placing one's finger on the end, and then pushing the plunger until it is physically unable to. When applied force on the plunger is released, the plunger springs back to its original position due to pressure. However, if that syringe is filled with water, the whole scenario becomes widely different, leading us into *Pascal's Law*, the direct mathematical foundation for hydraulics. But, first, let's analyze the physical properties of pressure and its relationship to energy.

2.1 Pressure

Pressure is defined as force per unit area, or:

$$P = \frac{F}{A} \quad (1)$$

and has the SI units Pascal, see figure 2, which are Newtons per square meter. It is often more convenient to use pressure rather than force to describe fluid behavior as they are more complicated systems.

For any object sitting on Earth, it experiences the force of gravity, and thus, the normal force exerted by the surface that it is resting on. That force of gravity is an object's weight, and the normal force is the feeling of having weight. However, for non-point masses, different orientations may yield different areas in contact with the surface, which yields different exerted pressures. Consider a 20 kg box with length 5 metres, width 10 meters, and height 2 meters. If you lay the box on its large side, the area in contact is 50 m^2 , and has a pressure computed out to:

$$P = \frac{200 \text{ N}}{50 \text{ m}^2} = 4 \text{ Pa}$$

But, if the box is laid on a more narrow side, the area in contact is 20 m^2 and the pressure then becomes 10 Pa . Intuitively, it makes sense why the pressure would be more. Standing on our feet, the pressure is much more manual, but if you try to balance on one finger, even just for a second, the pain on that hand becomes unbearable. When the area of contact decreases, the pressure increases. This explains why sharper objects cut easier with less amounts of force. When the tip of a blade is sharpened so that the area of contact is as small as possible, it cuts through materials like butter.

For non-solid objects, pressure is considered differently. For liquids, the medium is treated as a continuous distribution of matter, whereas a gas' pressure is considered to be the average pressure from molecular collisions with the wall. Think *Ideal Gas Law*: $PV = nRT$. When the number of moles increases, there are more gas molecules hitting the wall, increasing the pressure

Due to the nature of equation (1), adding another distance multiplier can change the nature

of the equation. Let's consider the box example again. Remember that work is defined as the force applied parallel to the displacement of an object, or:

$$W = F \cdot d \quad (2)$$

and that if adding an additional displacement modifier to area makes the unit m^3 , indicative of volume, where:

$$V = A \cdot d$$

then, one can substitute both of these equations into the numerator and denominator respectively for equation (1), so that:

$$P = \frac{F \cdot d}{A \cdot d} = \frac{W}{V} \quad (3)$$

where the displacement vector cancels. Then, by the definition of the third term in equation (3), pressure can be defined as a measure of the energy per unit volume of an object. Since energy is *stored* per unit volume when pressure is applied, that alludes to the fact that there is a kinetic and potential energy associated with fluids.

If pressure is able to be defined by the energy per unit volume of a liquid, then pressure should be able to be defined as kinetic energy per unit volume of a liquid, as well as potential energy per unit volume of a liquid. If a liquid is moving, then the standard kinetic energy equation for moving objects, being:

$$KE = \frac{1}{2}mv^2 \quad (4)$$

can be substituted into the W portion of equation (3), so that:

$$P = \frac{\frac{1}{2}mv^2}{V}$$

Remember that density is described by how much mass fits into some specific volume, or mathematically:

$$\rho = \frac{m}{V} \quad (5)$$

which can then be substituted in the prior equation where:

$$P = \frac{1}{2}\rho v^2 \quad (6)$$

Furthermore, one can define pressure in terms of potential energy per unit volume as well, which can be used by deriving the potential energy for a mass suspended midair. If work equals the force acting parallel to the displacement to an object, as seen in equation (2), and if intuitively, the object does fall some amount of height h , then the potential energy stored in that object is:

$$U = mg \cdot h \quad (7)$$

which can then be substituted into equation (3). The reason this is the case is because, technically, water molecules are subject to gravity and will tend to *fall*, thus, storing energy. With further consideration to the density equation (5), pressure is defined as:

$$P = \rho gh \quad (8)$$

This equation is also known as the hydrostatic pressure of a liquid, which is the pressure exerted by a fluid at rest or in equilibrium, caused by the force of gravity. It is the weight of a liquid pushing down on itself and all other walls that surround it, and explain why it is unsafe to go deep into the ocean because hydrostatic pressure becomes so high. It also explains why one's ears will pop if they dive into water. When going deeper into water, the hydrostatic pressure increases overhead, causing a subsequent equalizing reaction of the ears to balance pressure.

2.2 Pascal's Law



Figure 2: French mathematician, philosopher, physicist, and prose writer; discovered Pascal's Law [2].

Considering the syringe scenario again, when the syringe is instead filled with a liquid such as water, the fluid is not as compressible in comparison to air, and begins to exert its pressure forces on the exterior of the syringe. Pascal's law restates this phenomena, claiming that any pressure applied to a confined liquid has its value project equally and undiminished (does not weaken) in all directions to every part of the whole fluid and the walls of its container.

Imagine a champagne bottle, like one seen in figure 3, has a stopper applying a constant force to keep the champagne contained. Let's say that the force the stopper is exerting on the tip of



Figure 3: Pressure is applied due to the stopper pushing down on the bottle [3].

the bottle is equal to 10 N , and the area is 5 cm^2 . Therefore, with help from equation (1), the pressure at the tip is equal to $20,000\text{ Pa}$ or 20 kPa . With attention to Pascal's Principle, the pressure should then be exerted to every part of the liquid as well as the system that encloses it. Then, the sides of the bottle also experience a pressure equal to 20 kPa . What, though, is the force exerted on the sides of the bottle.

Let's consider the bottom of the bottle to have a pressure of 20 kPa with a surface area equal to 50 cm^2 . Rearranging equation (1) for force gives us:

$$F = PA \quad (9)$$

where the force at the bottom of the bottle is equal to 100 N . Wait, the stopper of the bottle was exerting a force of 10 N , how is it possible that there is more force at the bottom of the bottle than at the top. Remember that pressure goes undiminished, not force, meaning that there can be different forces at different parts of the bottle with the same pressure. Pascal actualized this through an experiment with a long narrow tube attached to a barrel. When the tube's height increased, the hydrostatic pressure increased as well, which was consistent with equation (8). However, as height increased, the barrel became more subject to bursting, until it did. Hence, Pascal realized that pressure was increasing throughout the entire system, but the force that was applied to the walls of the barrel changed.

Knowing that force becomes multiplied when pressure is the same for a container, but different for specific areas of the container, is the driving *force* for the concept of hydraulics. Moreover, it should be noted that at the bottom of any fluid container, there is some hydrostatic pressure overhead due to all the water molecules affected by gravitational potential energy. The pressure at the bottom of the container should then become:

$$P_{bottom} = P_{top} + \rho gh \quad (10)$$

2.3 Hydraulic Press

Now, we'll analyze a common hydraulic press scenario to illustrate the mathematics behind them. Consider an apparatus such as one seen in figure 4, where there is a hydraulic press

attempting to move a car. m_{car} , d_1 , and d_2 are given, and we are asked to evaluate what force F_{piston} is required to move the car a distance d_2 .

Due to the conservation of energy principle, we can assert that the energy input of the system is equal to the energy output of the system, with the neglecting of non-conservative forces, so that:

$$W_{in} = W_{out} \quad (11)$$

And, with the equation (2), this equation expands out to:

$$F_{piston} \cdot d_1 = m_{car}g \cdot d_2 \quad (12)$$

where the right-hand side is gained by the acknowledgement that the weight of the car is being moved by a distance d_2 . The dot products of the forces and distances can be replaced by standard multiplication since the force is acting directly parallel to the displacement, and moreover, the equation (12) can be manipulated to solve for F_{d_2} where:

$$F_{piston} = m_{car}g \frac{d_2}{d_1} \quad (13)$$

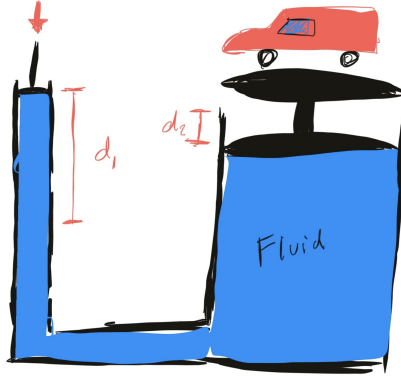


Figure 4: Hydraulic press lifting a car (excuse the crude drawing).

This introduces a very important fact about hydraulics, in that, if you want F_{piston} to be small (in order to export a large amount of force), you must make it so that d_1 is a large value. If we introduce some numbers and say that the car is a *2025 Toyota Camry* with a mass approximately 1600 kg , then, in order to move the car up 5 m , with d_1 of 30 m , the force applied to the first piston should equal 2613 N . Although this force is noticeably smaller than the weight of the car, it is still around the weight of a grizzly bear. However, if we make it so d_1 is instead 100 m , then the force applied to the first piston becomes equal to 784 N .

This is still not what we want! If we want hydraulics to be powerful, we should derive another way to decrease the force applied to make the *2025 Toyota Camry* to move upwards by 5 m . With that, we'll claim that r_1 and r_2 radii of the pistons respectively are given, and that the

pistons have the surface area conforming to the area of a circle. Due to the equation (1), there is a pressure exerted undiminished and equally to every part of the system by the car, where:

$$P_{car} = \frac{m_{car}g}{\pi r_2^2} \quad (14)$$

Then, there is some force being exerted on the surface area of piston #1 where:

$$F_{piston} = P_{car}\pi r_1^2$$

which after have its pressure term expanded due to equation (14), the force applied to the piston becomes:

$$F_{piston} = m_{car}g\frac{r_1^2}{r_2^2} \quad (15)$$

Let's lock in the F_{piston} in equation (15) to be 100 N , equivalent to lifting a large watermelon. Let's then revert the initial distances d_1 and d_2 to 30 m and 5 m respectively. Then, it seems intuitive to change r_1 and r_2 to get the intended 100 N I desire.

For incompressible fluid, any push in any direction must be followed by an equal and opposite push in a different direction. In other words, because there is no compressing, the mass of the fluid stays the same, and because the amount of fluid doesn't change, the volume of the fluid should not either. This concept is known as the conservation of volume, which holds that:

$$V_{in} = V_{out} \quad (16)$$

So, in this hydraulic press scenario, the volume changes that occur when a piston is pushing down are equal to the cross-sectional area times the displacement of the piston. If the first piston is pushed down, then the second piston is pushed up, which yields a corresponding volume change as well. Thus, using equation (16), we can see that both volume changes must equal each other to satisfy the conservation of volume, where:

$$\pi r_1^2 d_1 = \pi r_2^2 d_2 \quad (17)$$

which yields a non-linear equation where r_1 and r_2 must satisfy:

$$\frac{r_2^2}{r_1^2} = \frac{d_1}{d_2} \quad (18)$$

Now, that means there is some constraint to the situation, and that, we cannot get exactly 100 N pushing down on the piston. We will always get a F_{piston} that is consistent with a computation from the equation (13), and that the smallest possible force I can get is constrained by the distances the pistons travel by, which is actually a good conclusion to changing design constraints.

2.4 Hydraulic Brakes

Hydraulic brake calipers, like one seen in figure 5, use pressurized fluid to push internal pistons and force brake pads to clamp onto the spinning rotor driving the wheel. When the brake pads clamp on to the rotor, a friction force is applied directly opposite to the linear motion of the motor, and applies a torque corresponding to the lever arm it is applied on.

Whenever one presses a brake pedal in the car, there is a mechanical signal for an engine vacuum or auxiliary pump to multiply the force applied to a piston in a master cylinder filled with hydraulic fluid. The piston in the hydraulic fluid creates pressure equally and undiminished in all directions that travels in tubes that route to the brake caliper. Inside the caliper are small mechanical pistons attached to brake pads that then push inwards and apply force to the rotor. Let's reverse engineer the amount of force that the piston in the master cylinder is applying to achieve stoppage for a car traversing 60 *mph*. Considering the brilliant engineering of *Brembo* brakes, 60 to 0 stoppage can a very short amount of time. We can then derive the average force applied during that amount of time by utilizing the impulse equation (21). Impulse as the total change in momentum caused by a force applied over some specific amount of time. This comes from the characterization that force is defined as the change in momentum over the change in time, or:

$$F = \frac{dp}{dt} \quad (19)$$

which is intuitive as force causes acceleration, a change in velocity, with respect to the inertia of that object, or mass. And, since momentum is an object's velocity times its inertia, then a force will cause a change in that momentum. So, considering equation (19), we can solve for dp and get:

$$dp = F dt \quad (20)$$



Figure 5: *Brembo* is a world leader and pioneer in disc brake technology for the automotive industry [4].

From this equation, we can claim that F can be simplified by redefining it as the average force applied during dt , or, F_{avg} . Furthermore, dp and dt can be changed to more macroscopic

variants where:

$$\Delta p = F_{avg} \Delta t$$

where if p is expanded out to its definition, the above equation becomes:

$$m(v_{final} - v_{initial}) = F_{avg} \Delta t \quad (21)$$

assuming that the mass of the object does not change. Then, rearranging for F_{avg} yields:

$$F_{avg} = \frac{m(v_{final} - v_{initial})}{\Delta t} \quad (22)$$

Therefore, this becomes the amount of force needed to completely halt the car with some initial velocity. However, there is a caveat to this equation. F_{avg} is the force acting directly against linear motion, but is not the force that is acting upon the entire car. If this were true, then there would be one mega brake for the entire car, but each wheel usually has its own brake. Thus, we can use F_{avg} to develop a smaller braking force, specifically for the front wheels, by the equation:

$$F_{wheel} = \frac{\phi F_{avg}}{2} \quad (23)$$

where ϕ is the braking share between the front and back wheels which will be denoted by ϕ_{front} now, and the entire equation is divided by two since both front wheels endure the force and exert a braking force. Furthermore, we must introduce the torque τ which more accurately describes the situation at hand. When cars move, they are not only advancing further, but are having its wheels rotate to propel itself forward. Therefore, there must be some additional energy at hand to halt the car as well as to stop rotation of the wheels. Torque is the measure of an object's angular acceleration coupled with its inertia. However, the inertia for objects with rotational motion is not mass, but rather, the moment of inertia.

Moment of inertia is the measure of an object's resistance to changes in rotational speed. We can find the simple definition of moment of inertia by manipulating the kinetic energy equation (4). Rotational velocity can be related to linear velocity by claiming that linear dimensions are related to angular dimensions through the radius and angle. For example, rotational displacement θ times a radius r yields an arc length l . That arc length can be equated to linear displacement d if it is stretched out like a tangent line. An example is that when a wheel of a tire rotates, the corresponding distance traveled is equal to the arc length corresponding to the radius of the tire. So, this can be extrapolated to other forms of linear motion, including velocity, making the kinetic energy equation:

$$KE = \frac{1}{2} m r^2 \omega^2 \quad (24)$$

We can then take the term $m r^2$ and define it as I . For linear kinetic energy, the inertia term is m , but for angular energy it is I . Then we can think of these two quantities as the inertia term corresponding to the specific type of motion. However, I is not defined as $m r^2$, but rather, it is

defined as the inertia at every single radius pertaining to the object that is rotating. Therefore, the moment of inertia should be equal to the sum of inertias throughout the radius, or:

$$I = \int r^2 dm \quad (25)$$

Let's model the internal motor as a simple disk with mass m . The disk has an areal mass density of σ that is defined as:

$$\sigma = \frac{m}{A} \quad (26)$$

Then, we can rearrange the prior equation for m , and take the basic derivative of all components, such that:

$$dm = d\sigma dA \quad (27)$$

and substitute into equation (25) to get:

$$I = \int r^2 d\sigma dA \quad (28)$$

Then, we must evaluate dA in terms of r so we can integrate the r^2 term. The area of a circle is defined as $A = \pi r^2$, where the derivative becomes:

$$dA = 2\pi r dr \quad (29)$$

Let's also define the lower limit and the upper limit of the integral in equation (28) as 0 and r respectively, since we are evaluating the moment of inertia at every point of the circle from the center to its edge. This allows us to take the $d\sigma$ term out of the integral because the mass density does not vary by r , and the integral is from the center to the edge, containing the whole of σ . Finally, if we substitute the equation (29) into the updated integral we get:

$$I = 2\sigma\pi \int_0^r r^3 dr \quad (30)$$

Then, taking the integral yields:

$$I = \frac{1}{2}\sigma\pi r^4$$

Our work is still not done, though. Since areal mass density has a definition, given by equation (26), we can further simplify the above equation and get:

$$I = \frac{1}{2}mr^2 \quad (31)$$

Thus, we can claim that Newton's 2nd Law applies to situations pertaining to rotational momentum, where:

$$I\alpha = \Sigma\tau = T - F_{wheel}R \quad (32)$$

where R is the radius of the tire. The reason for including the second term on the right-hand side is that the road exerts an additional torque associated with moving forward. T is the term

that describes the braking torque exerted by the brake caliper. Furthermore, one can solve for T using equation (32) and yield:

$$T = F_{wheel}R + I\alpha \quad (33)$$

So, the braking force must satisfy the above equation in order to effectively stop the car. Let's denote the net torque of (32) as T for simplicity sake, and claim that the clamping force of the caliper is N , where it is pinching the internal rotor at a radius r_{pad} . Then, the equation relating the two quantities is as follows:

$$T = 2\mu_k N r_{pad} \quad (34)$$

where the force is doubled since clamping yields two forces acting on the same object, and μ_k is the friction coefficient associated with the rotor and the brake pads. Then, N can be isolated, T can be expanded via equation (32), and α can be written out as a rotational kinematic equation solving for angular acceleration to get:

$$N = \frac{F_{wheel}R + I\left(\frac{\omega_{final} - \omega_{initial}}{\Delta t}\right)}{2\mu_k r_{pad}}$$

and then, when F_{wheel} is expanded, as well as the moment of inertia, the equation yields:

$$N = \frac{\phi_{front}m_{car}(v_{final} - v_{initial})R + m_{rotor}r_{rotor}^2(\omega_{final} - \omega_{initial})}{4\mu_k r_{pad}\Delta t} \quad (35)$$

Then, using the pressure equation (1), we can claim that the pressure needed in the hydraulic fluid to exert a force N is:

$$P = \frac{N}{A_{brakes}} \quad (36)$$

where A_{brakes} is the surface area of the pistons associated with the brake pads. Furthermore, the force needed to create a pressure from the master cylinder should then be:

$$F = N \frac{A_{master}}{A_{brakes}}$$

where A_{master} is the surface area of the piston associated with the master cylinder of the car. Then, fully expanded out, the force required by the master cylinder to halt the car at some specific initial speed is:

$$F = \frac{\phi_{front}m_{car}(v_{final} - v_{initial})R + m_{rotor}r_{rotor}^2(\omega_{final} - \omega_{initial})}{4\mu_k r_{pad}\Delta t} \frac{A_{master}}{A_{brakes}} \quad (37)$$

Finally, we can consider some numerics to solve this qualitatively. Let's assume the *2025 Toyota Camry* situation again, this time modded with *Brembo* brakes. Because the car is Front-Wheel-Drive, its front wheels endure most of the braking force, where $\phi_{front} = 0.8$. The mass of the car and the internal rotor is 1600 kg and 10 kg respectively. The radii of the tires and internal rotor

are $0.33m$ and $0.15m$ respectively, whereas the radius of the brake pad is $0.125m$. Furthermore, $v_{initial}$ and $\omega_{initial}$ have values corresponding to the car moving at 60 mph , so their quantities are $26.8m/s$ and $81.2rad/s$ respectively. The kinetic friction coefficient associated between the brake pads and internal rotor is 0.4 . Finally, let's assume that A_{master} and A_{brakes} have circular pistons with radii r_{master} and r_{piston} with values $0.012m$ and $0.030m$ respectively. Area for a circular piston should correspond to the equation πr^2 . Finally, 60 to 0 stoppage for Brembo's happens quickly, and Δt can be considered to be $2.2s$. Then, the force exerted by the master cylinder is:

$$F = \frac{0.8(1600\text{ kg})(0 - 26.8\text{ m/s})(0.33\text{ m}) + 10\text{ kg}(0.15\text{ m})^2(0 - 81.2\text{ rad/s})}{4(0.4)(0.125\text{ m})(2.2\text{ s})} \frac{\pi(0.012\text{ m})^2}{\pi(0.030\text{ m})^2}$$

or,

$$F \approx 4120N$$

where the negative sign is negated because it is the braking force. If the vacuum booster and brake pedal multiply the force exerted by the foot on the brake pedal by around 8 times, then the force exerted by the foot in a panic situation is:

$$F_{foot} = 515N$$

which is a completely reasonable value, and concludes this section on hydraulic brakes.

2.5 Hydraulic Fluids

Water is a perfectly fine fluid to use for hydraulics since it is incompressible. However, in most if not all industrial applications, hydraulic oil is used, for three main reasons: temperature, lubrication, and corrosion. The subsequent sections will explain these consequences in more depth.

To preface, the hydraulic fluid used most commonly (90% of the time [5]) is mineral oils, see figure 6, will be referred to as a juxtaposition to water.



Figure 6: Mineral oil solutions are used most commonly in hydraulic situations [6].

2.5.1 Temperature

The problem with water pertaining to temperature is that many applications can see heats above the vaporizing temperature, or below the melting temperature. For example, a car's master cylinder will see temperatures in the range of $150\text{ }F^{\circ}$ to $220\text{ }F^{\circ}$, well above the vapor temperature of water. Thus, if water is used in this scenario, the water will spontaneously boil immediately, and there will be no fluid to compress making Pascal's law inapplicable. Gases are compressible because of the wide space between molecules. Since there is no pressure exerted equally and undiminished in all directions of the medium, the brake pistons are never able to see the pressure, and thus, not able to exert an outward force. For general applications, mineral based oils begin to boil at $288\text{ }F^{\circ}$, clearing the temperature of the master cylinder.

On the other end of the extreme, cold-based applications for hydraulics will see negative effects when using water. Construction equipment that utilizes hydraulic systems can freeze during the winter and cause machinery to break down. When water freezes into a lattice grid, its volume increases in certain sections which can apply intense stresses around its enclosure. It's analogous to placing soda in a freezer, and realizing hours later that the walls of the freezer are coated in sugar. Therefore, it is common to use mineral based oil for cold applications, which have specialized polymers that protect microscopic particles from spontaneously freezing. Some hydraulic fluids can withstand $-40\text{ }F^{\circ}$, which is around 6° colder than the normal temperature of an arctic winter. Sir William Armstrong, inventor of the hydraulic accumulator, hydraulic crane, and the *Armstrong* gun, realized the pain of water freezing in hydraulics, and attempted to include additives in the fluid like glycerol to lower the freezing point. However, this did not solve the problem of corrosion, see section 2.5.3, which would break down the surrounding material.

2.5.2 Lubrication

Water can increase friction between moving parts in two ways. One is its polarity. H_2O has a natural electrical polarity that causes it to want to bond to other molecules or atoms. Physics experiments show that this microscopic tendency of attraction causes an overall drag that increases friction between two objects. There is also a phenomena called the meniscus effect. Water has surface tension that forms a tight skin, if this surface tension is present between two close surfaces, the tight skin will create a resistance when those two surfaces attempt to move. Again, these are all microscopic effects, but when multiplied over thousands of times, it can cause a great loss in efficiency.

Mineral based hydraulic fluid contains hydrocarbons, or oils, that do not have polarity and have weaker surface tension. The reason for the latter is that surface tension for hydrocarbons is created by much weaker intermolecular forces, rather than powerful hydrogen bonds for water. Oils also have the effect of moving around dirt in a system, which causes unnecessary abrasiveness regarding moving parts. Because of its non-polar nature, it has the capacity to dissolve other non-polar molecules like grime and dirt surprisingly well, and carry them to a filter to remove them.

2.5.3 Corrosion

The most detrimental effect that water has when used in hydraulics is its oxidation reaction with iron. The way rust forms from iron particles and oxygen particles is from the chemical equation:



Naturally, iron atoms will shave off a steel surface due to the oxygen atoms in H_2O having high polarity and wanting to attach due to high electronegativity. The main problem with rust is that it will flake off over time revealing more layers of fresh iron. This reaction compounds until fractures occur from sections of the steel being smaller and weaker than others, causing huge mechanical failures especially with systems associated moving parts. Hydrocarbons do not garner the same effect because there are no water molecules present to give iron atoms the necessary oxygen atoms to form iron oxide. Furthermore, when a system is supplied with oil, the air molecules outside of the system are effectively sealed off from entering, meaning that no auxiliary oxygen atoms or moisture can enter to produce iron oxide. If one ever tries to mix oil and water, it is clear that oil is extremely hydrophobic, repelling the water due to its non-polar, non-reactive, nature.

3 Components, Explanations, and Examples



Figure 7: Missile launcher (not hydraulic launcher, but still looks cool) [7].

Now, let's turn our attention back to the subject of hydraulic launchers. This section will detail some key components of a hydraulic launcher, which will inevitably be pivotal to the design and fabrication of the physical product; it serves as a roadmap for choosing and selecting certain components to include in the design. I'll go through each component, explain what they do, and then give various examples of them in action. It is not exactly clear how the hydraulic launcher will look from this section, but it serves as a basis for the skeleton of one.

3.1 Power Supply

A Hydraulic Power Unit (HPU) is the system of interconnected pumps, filters, and valves transmitting fluid flow that converts mechanical energy into hydraulic energy [8]. This conversion can actually create more power and force than normal electrical/mechanical units. The underlying path of an HPU is the engine to the pump, to the hydraulic cylinder, and back into a reservoir. The engine for an HPU can either be an electrical motor or an internal combustion engine, and it connects to a rotary pump that traps a fixed volume of fluid and forces it from the inlet to the outlet. Figure 8 shows the process by which it is commonly done by. This motor is also

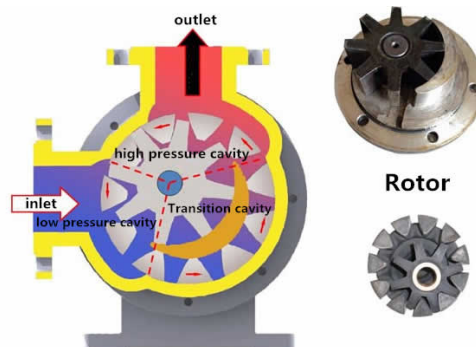


Figure 8: Schematic showing how rotary pumps transport fluid from the inlet to the outlet [9].

analogous to the hydraulic pump, that pushes the liquid from a reservoir, essentially storage of hydraulic fluid, towards the hydraulic cylinder which converts the fluid pressure, created by a different component of the system, into mechanical power. The hydraulic cylinder is considered to be the last step in the entire process. This is extremely similar to the hydraulic braking example in section 2.4 where the master cylinder produces an intense amount of pressure which is applied to the brakes for each wheel. The actual components of the Hydraulic Power Unit that should be analyzed in this section is the engine and the rotary pump that is attached to the motor shaft. We can gain the specified torque output from motor catalogs to calculate the amount of flow that is supplied to the hydraulic system. HPUs, like one seen in figure 9, will often have an accompanying data-sheet that indicates the flow rate of the HPU, and the torque and/or angular speed of the electric motor. The maximum shaft speed N_{max} that is given for the HPU in 9 is $1,725\ rpm$ which can be converted to rad/s via the following equation:

$$\omega = \frac{2\pi N_{max}}{60\ s} = 180.6\ rad/s \quad (39)$$

Power is the measurement of how much work is done over some specific amount of time where:

$$P = \frac{W}{t} \quad (40)$$

Since work is the energy created from force exerted in the direction of displacement from equation (2), and if torque equals the force F applied perpendicularly to the moment arm r , or:

$$\tau = F \times r \quad (41)$$

and r can be thought of as the radius of the point where force is applied to the axis of rotation. If torque is multiplied with angular velocity ω , the following relations can be made:

$$F \perp r \cdot \omega = F \perp v = F \cdot \frac{\text{displacement}}{\text{time}} = \frac{W}{t} = P \quad (42)$$

$$\tau \cdot \omega = P \quad (43)$$

$$\tau = \frac{P}{\omega} \quad (44)$$

Therefore, using rated wattage from a data sheet, and specifically for the motor in figure 9, the maximum torque output is:

$$\tau_{rated} = \frac{(1\ hp = 735.5\ W)}{180.6\ rad/s} \approx 4.1\ Nm \quad (45)$$

This is slightly alluded to in equation (42), where we can gain work from torque times angular displacement. This is seen through the multiplication of ω which is angular displacement θ over time t . Therefore, just using the θ term will yield the equation:

$$W = \tau\theta \quad (46)$$

Therefore, for one full rotation of the motor shaft, the work done is:

$$W_{shaft} = 2\pi\tau_{pump}$$

where τ_{pump} corresponds to the value needed for one full rotation, and 2π is the value in radians for one full rotation. Furthermore, we can introduce a concept called flow work, or rather, the energy needed to maintain a pressure differential and to also maintain the flow of fluid at a given velocity. If velocity changes, the kinetic energy also changes, which means that work also must be constant in this scenario. Flow rate is defined by the equation:

$$W = \Delta PV \quad (47)$$

where ΔP is $(P_{out} - P_{in})$, and is referred to as the pressure differential. The pump displaces a fixed amount of volume V for every revolution. For instance, a rotary pump like one in figure 8, will flush out a certain amount of volume granted that the reservoir is properly supplying the hydraulic system with fluid. Furthermore, this fixed volume will be denoted as D which is the specific volume flushed from the inlet to the outlet in one revolution. We can then also equate W in the above equation to W_{shaft} where:

$$2\pi\tau_{pump} = \Delta PD \quad (48)$$

This equation holds for an ideal pump, where there are no losses due to friction as the pump rotates. However, we can account for mechanical losses with the term η_m , where τ_{actual} can be empirically measured so that:

$$\eta_m = \frac{\tau_{ideal}}{\tau_{actual}} \quad (49)$$

τ_{actual} should always be greater τ_{ideal} because it requires more torque to rotate the rotary pump when non-conservative forces are present, therefore $\eta_m < 1$. The typical efficiency for AC induction engines is 0.75%. Then, the actual torque required to rotate the pump for one revolution, as in equation (48), should be:

$$\tau_{pump} = \frac{\Delta PD}{2\pi\eta_m} \quad (50)$$

What equation (50) essentially means is that to maintain the pressure differential of ΔP for a rated pump displacement of D , for one full revolution of the pump, the torque the pump needs must clear τ_{pump} while accounting for mechanical losses due to friction. Furthermore, if the maximum rated torque is substituted into equation (50), we can solve for the maximum possible pressure differential that the hydraulic power unit can maintain, where:

$$\Delta P_{max} = \frac{2\pi\eta_m\tau_{rated}}{D} \quad (51)$$

We can use the data sheet for the hydraulic power unit corresponding to figure 9 to obtain the flow rate Q that is equal to 0.9 *gallons/min*. Although this contains the volume unit, we need

to do three things: convert the gallons into SI units, find pump displacement per revolution, and remove the time unit from the flow rate. Frequency is defined as the amount of revolutions per second. We can use the angular velocity ω of the pump and find the frequency of the rotary pump by 2π radians to get revolutions per second, or frequency, where:

$$f = \frac{\omega}{2\pi} \text{ rev/s} \quad (52)$$

Then, all that is required is converting gallons to m^3 where $1 \text{ g} = 0.00378541 \text{ m}^3$. Then, using flow rate and frequency, we can find pump displacement per revolution via the equation:

$$D = \frac{Q}{f} \quad (53)$$

Then using units, we find that:

$$D = \frac{0.00005678116 \text{ m}^3/\text{s}}{\frac{180.6 \text{ rad/s}}{2\pi}} = 0.00000197568 \text{ m}^3/\text{rev}$$

But, as always in engineering problems, there is some discussion to be had about efficiency. η_v represents the volumetric efficiency of the rotary pump, and moreover, the ratio between Q_{actual} and Q_{ideal} , or:

$$\eta_v = \frac{Q_{actual}}{Q_{ideal}} \quad (54)$$

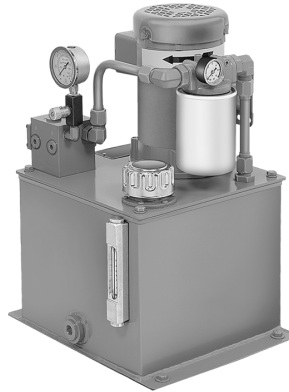


Figure 9: Picture of what a Hydraulic Power Unit can look like [10].

Since Q_{actual} is expected to be less than Q_{ideal} , η_v is less than 1. For the sake of numerics, let's denote η_v as 0.9, retrieving a new displacement rate that accounts for the volumetric efficiency of the rotary pump to be $D = 0.0000021952 \text{ m}^3/\text{rev}$. Finally, using the equation (??) and the

new displacement rate D , we can solve for the maximum pressure differential given the rated torque of the hydraulic power unit, where:

$$P_{max} = \frac{2\pi 0.754.1 Nm}{0.0000021952 m^3/rev} = 8801382.5 Pa = 8.80 MPa$$

There are two types of HPUs: *Single-Acting* and *Double-Acting*. The single-acting HPU has only pressing power, meaning the piston that applies mechanical energy is forced in one direction. They are often found in internal combustion engines and pumps, because only one range of motion is required (i.e. a hydraulic press needeth only to crush). Furthermore, as it is expected, a double-acting HPU has pressing and pulling power, two ranges of motion. A good example of a system that needs double-acting is an excavator. An excavator needs to apply force in scooping as well as in pushing. This is accomplished by the hydraulic cylinder having two ports where liquid can enter from, and a piston rod that is double-sided. HPUs also have variation in size, that being, micro power pack units, mini power pack units, standard hydraulic power pack units, and hydraulic unit stations. For industrial applications, the standard hydraulic power pack unit is housed inside the system. Using the excavator example again, it is found internally, located near the base, and must have pressurized hydraulic fluid travel up for arm and boom actuation.

3.2 Energy Storage

The energy reserve for a hydraulic launcher is known as the hydraulic accumulator. The accumulator is essentially a pressure storage device that holds a volume of pressurized hydraulic fluid so that it can supply energy, smooth out changes to pressure, and absorb system shocks in a hydraulic circuit. The accumulator in the hydraulic launcher will store energy in the event that large amounts of energy are required, that is, to launch an object. It is a blunder to think that the accumulator stores *hydraulic* energy, or hydrostatic energy, but it actually converts hydraulic pressure into potential energy through the compression of a gas, spring, or a raised weight. For each case, the weight being raised created gravitational potential energy, the compressed spring creates spring potential energy, or $1/2(kx^2)$, and the compressed gas stores pressure potential energy. Figure 10 shows two different types of accumulators. The bladder accumulator utilizes a flexible bladder, where the pack is filled with nitrogen gas through a valve to a set pressure. When the overall pressure of the system rises, hydraulic fluid enters the steel vessel and applies force to the surface area associated with the bladder, compressing it. When system demand increases, in the event that large hydraulic energy is needed, the compressed air expands and forces the fluid back into the hydraulic circuit, which then applies backwards pressure on the fluid and creates the large release of energy. Bladder accumulators offer low inertia for systems with rapid work cycles and high-frequency pulsations due to the elastic nature of the bladder pack. However, bladder accumulators lack in their ability to handle high compression ratios due to the tradeoff of having a more flexible storage unit. Piston accumulators operate in the exact same way that bladder accumulators do, where the chamber

is initially filled to a certain pressure, which then compresses from the hydraulic circuit until large releases of energy is needed. Due to the rigid design of the piston, larger gas compression ratios are permitted. Also, any mounting position can be used for a piston accumulator since the piston pushes liquid out, whereas bladder accumulators may have the problem of hydraulic fluid getting stuck in small crevices between the shell and bladder.

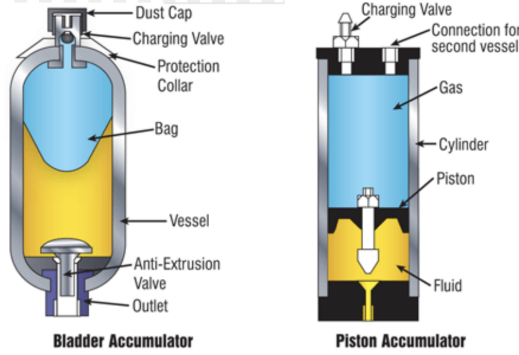


Figure 10: Schematic showing two different types of hydraulic accumulators [11].

Nitrogen follows a polytropic process where pressure and volume relate by the equation:

$$PV^n = C \quad (55)$$

where P is the pressure, V is the volume, and n is the polytropic index. P and V are relatively easy concepts to understand, but the latter term should be discussed further. The polytropic index n is a numerical exponent that defines how pressure, volume, and temperature relate during the expansion or compression of a gas. The polytropic number can be found if the final and initial states of a gas are known during a compression/expansion process, where:

$$P_i V_i^n = P_f V_f^n$$

then, the equation is rearranged by taking the natural log of both sides to yield:

$$n(\ln(\frac{V_i}{V_f})) = \ln(\frac{P_f}{P_i})$$

or finally:

$$n = \frac{\ln(P_f/P_i)}{\ln(V_i/V_f)} \quad (56)$$

Then, the number of n reveals what type of physical process is underway, that being, $n = 0$ indicating a constant pressure situation, $n = 1$ indicating a constant temperature situation, $n = C_p/C_v$, where C_p is the heat capacity at constant pressure and C_v is the heat capacity at constant volume, indicating an adiabatic situation, and $n = \infty$ indicating a constant volume scenario. An adiabatic situation is especially important because in the cases of launches, the process cannot occur isothermally since heat having no time to escape to other parts of the

simple. Therefore, the polytropic process of nitrogen in an accumulator should have a value of n falling between 1 and C_p/C_v . For diatomic ideal gases (nitrogen occurs as N_2), heat capacity for C_v and C_p is $(5/2)R$ and $(7/2)R$ respectively, making $n = 1.4$ for adiabatic scenarios. Unfortunately, though, at high temperatures, nitrogen does not behave as a real gas and is better described by the *Benedict-Webb-Rubin Equation*.

Let's consider that we have an initial volume V_0 and an initial pressure P_0 , if we use equation (55), we can equate initial pressure and volume to future volumes and pressure. Let's also hold that the process happens isothermally, which it won't, but for the sake of simplicity. This takes the polytropic index exponent out of the equation, where:

$$P_0 V_0 = C \quad (57)$$

Remember that work is done when a force is exerted on an object and it displaces. Whenever there are volume changes for a system with pressure, displacement is happening for the vessels holding the ideal gas. However, with changes of volume corresponding to negative values, there are corresponding changes to displacement in the direction to the system. Since work done on the system is positive, the equation for work is:

$$W = -P\Delta V \quad (58)$$

Furthermore, if we want to get an accurate reading of the total work that is produced when there are changes in pressure, we should analyze every little change in volume dV where infinitesimal changes in pressure happen to get the total work:

$$W = - \int P dV \quad (59)$$

Using equation (55), we find that the pressure is C/V when n is equal to 1. Furthermore, substituting into the aforementioned equation yields:

$$W = - \int \frac{C}{V} dV \quad (60)$$

Since we are analyzing the energy created when there is a change in volume, when the nitrogen gas is compressed vs. expanded, the integral limits should be from V_1 to V_2 . V_1 represents the volume after hydraulic fluid compresses the gas, either in piston or bladder accumulators, and V_2 represents the volume when the gas chamber expands somehow. Then:

$$W = - \int_{V_1}^{V_2} \frac{C}{V} dV \quad (61)$$

after integrating, there will be the term V_2/V_1 which can be equated to P_1/P_2 , so that:

$$\frac{V_2}{V_1} = \frac{P_1}{P_2}$$

Also, using equation (57), we can claim that the constant in the equation is equal to the initial volume times the initial pressure, finally making equation (60):

$$W = -P_0 V_0 \ln\left(\frac{P_1}{P_2}\right) \quad \text{or} \quad P_0 V_0 \ln\left(\frac{P_2}{P_1}\right) \quad (62)$$

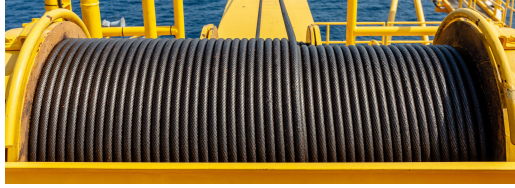


Figure 11: Heavy-duty cable winch used for lifting [12].

Which tells us that the work, or energy created, after a change in pressure, is equal to the initial pressure times the initial volume times the natural log of the ratio of the pressure after expansion and the pressure during compression. We can also use the ideal gas law equation (55) to find the accumulator volume V_0 for a corresponding change in pressure P_2 to P_1 that will create the energy as in equation (62). Remember in equations (55) and (57) where both sides can be equated where:

$$P_0 V_0^n = P V^n \quad (63)$$

where n is included for generality. This equation holds that for any pressure and any volume is valid so long that it satisfied the left-hand side. So, if we take the exponent of the inverse of n for both sides of the equation, the equation becomes:

$$P_0^{1/n} V_0 = P^{1/n} V$$

where after isolating V yields:

$$V = V_0 \left(\frac{P_0}{P}\right)^{1/n} \quad (64)$$

we can then extrapolate *any* V and *any* P to be the change in V and the change in P respectively, where for subscripts 1 and 2 denoting different V 's and P 's, we get:

$$V_2 - V_1 = V_0 \left(\frac{P_0}{(P_2 - P_1)}\right)^{1/n}$$

where equating $V_2 - V_1$ to ΔV , and expanding the fraction on the right-hand side yields:

$$\Delta V = V_0 \left(\frac{P_0}{P_2} - \frac{P_0}{P_1}\right)^{1/n} \quad (65)$$

then, finally, solving for V_0 , the volume that the accumulator should initially is:

$$V_0 = \frac{\Delta V}{\left(\frac{P_0}{P_2} - \frac{P_0}{P_1}\right)^{1/n}} \quad (66)$$

concluding this mathematical analysis on hydraulic accumulators.

A great example, especially for this project, are rollercoaster hydraulic launchers. Hydraulic oil and nitrogen gas are separately pumped into two sides of a piston. When the launch valves open, compressed nitrogen pushes hydraulic fluid through motors that spin a cable winch, see figure 11 to viciously accelerate the train car.

3.3 Valves

Hydraulic Circuits are the systems of interconnected components that use pressurized fluid to transmit energy through pressure, we already know the latter. But circuits allude to a different type of meaning. Generally, when we think of circuits in engineering, we imagine electronics, where current is charged by a battery/voltage and that interacts with different electrical components, where there are different ways to order components, in series or in parallel. Hydraulic circuits act the same way: they track the flow of fluid all throughout the entire system. An example is shown in figure 12 where the *flow* of hydraulic fluid can be seen for the whole system, as well as certain points. There are lots of different symbols on this diagram, but the most pertinent to this subsection are the squares, the valves.

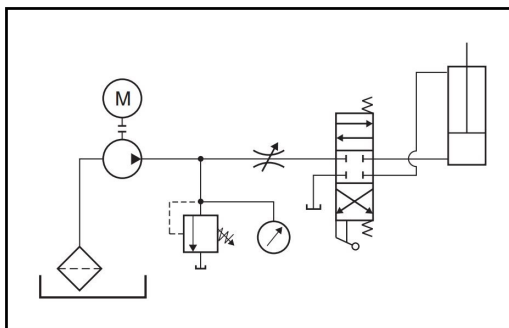


Figure 12: Schematic of a hydraulic circuit [13].

Considering that hydraulic circuits track the *flow* of the hydraulic fluid throughout the circuit, it is important to recognize that valves control the flow. A good example is an on/off valve for a hose. When the valve is fully tightened, the water cannot progress through the hose, and water does not spurt out of the spout. Loosening the valve yields a burst of water, exactly demonstrating how valves control the flow of fluid. The valves in the hydraulic launcher will be of the solenoid type, seen in figure 13. The metal plunger is wrapped in a coil of wire that is electrically charged and creates a magnetic field that immediately forces the plunger down to stop the flow of fluid (think of trying to connect two magnets of the same polarity). These are binary components meaning they can either be powered on or powered off, the supplying voltage is usually in the range of $12 - 48\text{ V}$. Solenoid valves often also have a certain pressure rating at lower temperatures. Because high temperatures weaken the bonds of materials, the solenoid plunger is vulnerable at high temperatures, so the pressure rating accounts for the maximum recommended pressure. However, all valves do not operate in the same way, and this

section outlines the different purposes that valves can have, and how they fit into a hydraulic circuit.

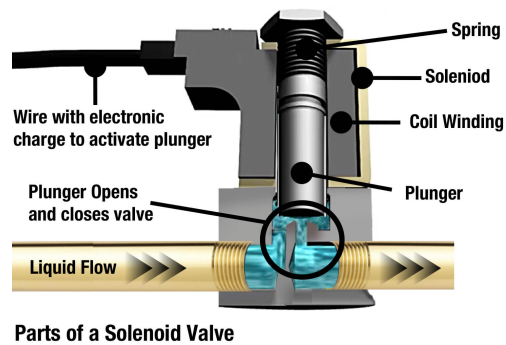


Figure 13: Labeling of different components for a solenoid valve [14].

3.3.1 Direction Control Valves (DCVs)

Directional Control Valves control direction, pretty counterintuitive.

3.3.2 Pressure Control Valves (PCVs)**3.3.3 Flow Control Valves (FCVs)****3.4 Actuation****3.5 Transmission****3.6 Recovery****3.7 Plumbing****3.8 Conditioning****3.8.1 Reservoir****3.8.2 Filtration****3.8.3 Cooling****3.8.4 Fluid****3.9 Sensors****3.9.1 Pressure Transducer****3.9.2 Temperature****3.9.3 Linear Motion Sensing****3.10 Control****3.10.1 Safety**

4 Regenerative Braking

4.1 Principles

4.2 Applications

4.3 Energy Storage

4.4 Components

4.5 Circuits

4.6 Dynamics and Losses

4.7 Measurement

5 Constraints

6 Approaches

7 Analysis

8 URCA

8.1 Abstract

8.2 Project Plan

8.3 Budget

8.4 Timeline

9 Conclusion

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